

DATA 266 — Homework 7

Generative Adversarial Networks and Autoencoders

I. Introduction

The main objective of this assignment was to understand Generative Adversarial Networks (GANs) and implement two types of autoencoders using PyTorch. The theoretical part focused on GAN architecture, mode collapse, and different GAN variants. The practical part involved training a Fully Connected Autoencoder (FC-AE) and a Variational Autoencoder (VAE) on the FashionMNIST dataset and comparing their reconstruction performance.

II. GAN Concepts

A GAN consists of two neural networks: a generator and a discriminator. The generator creates synthetic images, while the discriminator tries to identify whether an image is real or generated. Both networks learn through competition, with the goal of producing realistic images.

One common challenge in GAN training is mode collapse, where the generator repeatedly produces similar outputs instead of learning the full variety of the dataset. Techniques such as minibatch discrimination and WGAN-GP can help reduce this problem.

Three important GAN variants are DCGAN, Conditional GAN, and CycleGAN. DCGAN introduces convolutional architectures for image generation, Conditional GAN supports controlled generation using labels, and CycleGAN enables image translation between unpaired domains.

III. Dataset and Methodology

The FashionMNIST dataset contains 60,000 training images and 10,000 test images. Each image is a 28×28 grayscale image representing a clothing item. The images were converted into tensors with pixel values between 0 and 1.

Two models were implemented:

Fully Connected Autoencoder (FC-AE): The encoder compresses the input image into a 32-dimensional latent representation, and the decoder reconstructs the original image. Mean Squared Error (MSE) was used as the training loss.

Variational Autoencoder (VAE): The encoder learns the mean and log variance of a latent distribution. Using the reparameterization trick, the model samples latent vectors and reconstructs images. Training uses Binary Cross-Entropy (BCE) combined with KL divergence.

Both models used the Adam optimizer with a learning rate of 0.001, batch size of 128, and 15 training epochs. A fixed random seed of 1598 was used for reproducibility.

IV. Experimental Results

Both models completed 15 epochs of training. Their results are summarized below.

Metric	FC-AE	VAE
Trainable parameters	222,384	224,464
Training epochs	15	15
Final training loss	0.013095	243.9074
Test reconstruction MSE	0.013123	0.017353
Training time	191.43 s	208.80 s

Note: The VAE reconstruction MSE above uses the latent mean for deterministic evaluation. Its MSE using sampled latent vectors was 0.018722. The models' training losses are not directly comparable because they use different objectives.

V. Results and Discussion

The FC-AE showed a consistent decrease in training and test reconstruction loss. Its training MSE decreased from 0.044210 to 0.013095, while its final test MSE reached 0.013123. The training and test curves remained close, suggesting reasonable generalization without obvious overfitting.

The VAE also showed improvement throughout training. Its total training loss decreased from 312.7631 to 243.9074, and the sampled test reconstruction MSE decreased from 0.032589 to 0.018722. The KL divergence component helped regularize the latent distribution.

The FC-AE achieved lower test reconstruction error than the VAE in this experiment. This difference is understandable because the FC-AE directly focuses on reconstructing the input pixels, whereas the VAE balances reconstruction quality with learning a regularized latent distribution.

The notebook also includes reconstructed FashionMNIST images and newly generated images from the VAE. These visualizations help examine the models beyond numerical loss measurements.

VI. Conclusion

This assignment helped me understand how GANs and autoencoders approach generative learning differently. GANs learn through competition between a generator and discriminator, while autoencoders learn compressed representations of input data.

From the experiments, both FC-AE and VAE successfully learned to reduce reconstruction error. The FC-AE achieved better pixel-level reconstruction accuracy, with a test MSE of 0.013123 compared with 0.017353 for the VAE. However, the VAE also provides a probabilistic latent representation that supports generating new images.

Overall, the experiment demonstrated the trade-off between reconstruction accuracy and generative modeling, while providing practical experience implementing and evaluating neural networks in PyTorch.

VII. REFERENCES

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